1. TemporalEncoder (The Historian)

- **The Job:** To understand long-term historical trends.
- **How it works:** It uses 1D Convolutions (nn.Conv1d) to "slide" across your Weekly and Daily time windows. Instead of looking at the whole week at once, it looks at small chunks of time to figure out local patterns (e.g., "traffic usually builds up between 7 AM and 9 AM on weekdays"). Your final model actually creates two of these: one dedicated to analyzing last week's data, and one for yesterday's data.

2. SpatialEmbedding (The Translator)

- **The Job:** To prep the raw, recent traffic data for the complex spatial networks.
- **How it works:** Raw traffic speeds are just 1-dimensional numbers. This module uses a simple 1D Convolution to expand that single number into a 16-dimensional "feature vector" (SPATIAL_EMBED_DIM). Think of it like taking a flat sketch and turning it into a 3D model so the next layers have more angles to analyze.

3. GlobalSpatialEncoder (The Bird's-Eye View)

- **The Job:** To find invisible correlations across the entire city.
- **How it works:** We broke this down earlier! It uses Transformer-style **Multi-head Self-Attention** to look at all 325 sensors simultaneously. It mathematically figures out if a traffic jam at Sensor 10 is causing a ripple effect at Sensor 300, even if they aren't physically connected by a direct road in your dataset.

4. LocalSpatialEncoder (The Map Reader)

- **The Job:** To track traffic as it flows down the physical streets.
- **How it works:** This uses a Graph Convolutional Network (ManualChebConv). Unlike the Global Encoder, this module relies on your Adjacency Matrix (adj_mx). It specifically calculates how traffic speed at Sensor A will spill over into Sensor B based on the actual, physical road connections and distances between them.

5. TimeEncoder (The Calendar)

- **The Job:** To ground the model's predictions in reality.
- **How it works:** Traffic behaves completely differently on a Tuesday at 8:00 AM versus a Sunday at 2:00 AM. This module is a Multi-Layer Perceptron (MLP) that takes the exact Day of the Week and Hour of the Day (converted into a math-friendly format called "one-hot encoding") and injects that context directly into the traffic features.

6. TemporalDecoder (The Fortune Teller)

- **The Job:** To generate the actual future predictions.
- **How it works:** This is the heaviest part of your model. It takes all the combined intelligence from the previous modules (the historical trends, the global map, the local map, and the current time) and passes it through Transformer Attention layers. It outputs a specific sequence of predictions—in your case, the traffic speeds for exactly 1 hour, 2 hours, and 3 hours into the future.

- *(Note: It also uses PositionalEncoding so it doesn't get confused about which prediction belongs to which future hour).*

7. LSTGAN (The Boss / Orchestrator)

- **The Job:** To assemble the team and manage the assembly line.
- **How it works:** This is the master class. It doesn't do any math itself; instead, it initializes Modules 1 through 6. When a batch of data arrives, this class routes the Weekly data to the Historian, the current data to the Map Readers, synchronizes their outputs so they are all the exact same size (_match_time), glues them all together (torch.cat), and hands the massive final report to the Fortune Teller (Decoder).